Top 30 Most Asked

Python Interview Q&A 2025

To Crack any Interview

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1. What is Python?

Python is a high-level, interpreted programming language known for its clear syntax and readability. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming. Python is widely used in data science, web development, automation, and artificial intelligence.

2. What are Python's key features?

Python features include simple and readable syntax, dynamic typing, automatic memory management, a vast standard library, extensive community support, and portability across platforms. It is interpreted and supports both object-oriented and functional programming.

3. What is PEP 8?

PEP 8 is the official style guide for writing Python code. It provides conventions for code layout, naming conventions, line spacing, and more, to make Python code more readable and consistent across the community.

4. What is the difference between list and tuple?

Lists are mutable, meaning their elements can be changed after creation. They are defined using square brackets []. Tuples are immutable, meaning their elements cannot be changed, and are defined using parentheses ().

5. What are Python's data types?

Python supports various data types such as numeric types (int, float, complex), sequence types (list, tuple, range), text type (str), set types (set, frozenset), mapping type (dict), boolean type (bool), and special type (NoneType).

6. What is a dictionary in Python?

A dictionary is an unordered collection of key-value pairs. Keys must be unique and immutable. Dictionaries are defined using curly braces {}. Example: {'name': 'Alice', 'age': 25}.

7. What are *args and **kwargs?

*args allows a function to accept any number of positional arguments. **kwargs allows a function to accept any number of keyword arguments. They are used to create flexible functions that handle varying inputs.

8. What is a lambda function?

A lambda function is a small anonymous function defined using the lambda keyword. It can have any number of arguments but only one expression. It's often used for short, simple operations.

9. What is the difference between 'is' and '=='?

'==' compares the values of two variables, checking for equality. 'is' checks whether two variables refer to the same object in memory.

10. What is list comprehension?

List comprehension is a concise way to create lists using a single line of code. Syntax: [expression for item in iterable if condition]. It's more efficient and readable than using loops.

11. What is a Python module?

A module is a file containing Python code such as functions, classes, or variables that can be reused in other programs. Modules help in organizing code into manageable sections and are imported using the `import` keyword.

12. What is a package in Python?

A package is a collection of Python modules grouped together in a directory containing an `__init__.py` file. Packages help in structuring large Python projects and organizing related modules together.

13. What is the difference between deep copy and shallow copy?

A shallow copy creates a new object but inserts references into it to the objects found in the original. A deep copy creates a new object and recursively adds copies of the objects found in the original, using the `copy` module.

14. What are Python decorators?

Decorators are functions that modify the behavior of other functions or methods. They allow the addition of functionality to existing code in a clean, readable way. Defined using `@decorator_name` syntax.

15. What is recursion in Python?

Recursion is a technique in which a function calls itself to solve smaller instances of a problem. It continues until a base condition is met. Useful for tasks like calculating factorial, traversing trees, etc.

16. What is the purpose of 'self' in Python?

'self' represents the instance of the class and is used to access variables and methods associated with the object. It must be explicitly included as the first parameter in instance methods.

17. What is __init__ in Python?

`__init__` is a special method called a constructor that is automatically invoked when a new object of a class is created. It initializes object properties.

18. What are Python exceptions?

Exceptions are errors detected during execution. Python provides built-in exceptions like `TypeError`, `ValueError`, etc., and allows creating custom exceptions. They are managed using try-except blocks.

19. How do you handle exceptions in Python?

Exceptions are handled using try-except blocks. Optionally, `finally` can be used to execute code regardless of exceptions, and `else` can run if no exception occurs.

20. What is the use of the 'with' statement?

The 'with' statement simplifies exception handling for resources like files. It ensures that resources are properly cleaned up, for example, closing a file after opening.

21. What are Python generators?

Generators are special functions that yield values one at a time using the `yield` keyword. They are memory-efficient and used to generate large sequences on the fly.

22. What is the difference between global and nonlocal?

'global' allows modification of variables in the global scope, while 'nonlocal' allows access to variables in the nearest enclosing scope that is not global.

23. What are Python's memory management features?

Python uses automatic memory management that includes private heap space and garbage collection. The memory manager allocates memory and reclaims it when no longer needed.

24. What is the difference between mutable and immutable types?

Mutable types like lists and dictionaries can be changed after creation. Immutable types like strings and tuples cannot be modified after they are created.

25. What is duck typing in Python?

Duck typing is a concept where the type or class of an object is less important than the methods it defines. If it behaves like a duck (has required methods), it's treated as one.

26. What are magic methods in Python?

Magic methods are special methods with double underscores, like `__init__`, `__str__`, `__len__`, used to define custom behavior for built-in operations.

27. What is the use of the pass statement?

`pass` is a null statement used when a statement is syntactically required but no code needs to be executed.

Useful in defining empty functions or loops.

28. What is slicing in Python?

Slicing allows extracting parts of sequences like lists, strings, or tuples using the syntax `sequence[start:stop:step]`. It helps in accessing subsets of data.

29. What are iterators and iterables?

An iterable is any Python object capable of returning its members one at a time, like lists or strings. An iterator is an object that implements the `__next__()` method to return elements.

30. What is the difference between range and xrange?

In Python 2, `range` returns a list, while `xrange` returns a generator-like object for better memory efficiency. In Python 3, only `range` is available, behaving like `xrange`.